

Adversarial Example Does Good: Preventing Painting Imitation from Diffusion Models via Adversarial Examples

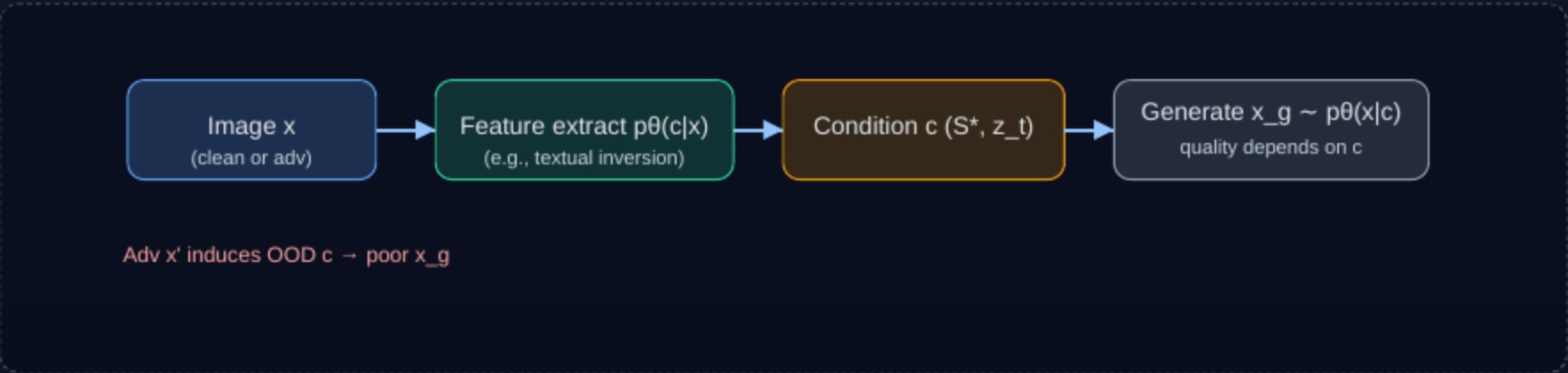
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Introduction & Motivation

- Latent Diffusion Models (LDMs) excel in AI-for-Art but raise copyright concerns.
- Textual inversion enables style/content imitation from few samples.
- Artists lack technical tools to prevent unauthorized training or imitation.
- Goal: use adversarial examples to hinder diffusion models from extracting features.



Abstract — Key Ideas

- They define adversarial examples for diffusion models and align evaluation with the DM inference workflow.
- They introduce AdvDM: Monte Carlo optimization over reverse-process latents to craft perturbations.
- They show that adversarial examples hinder feature extraction in diffusion models while preserving semantics.
- They report extensive experiments validating effectiveness on LDMs and real scenarios.

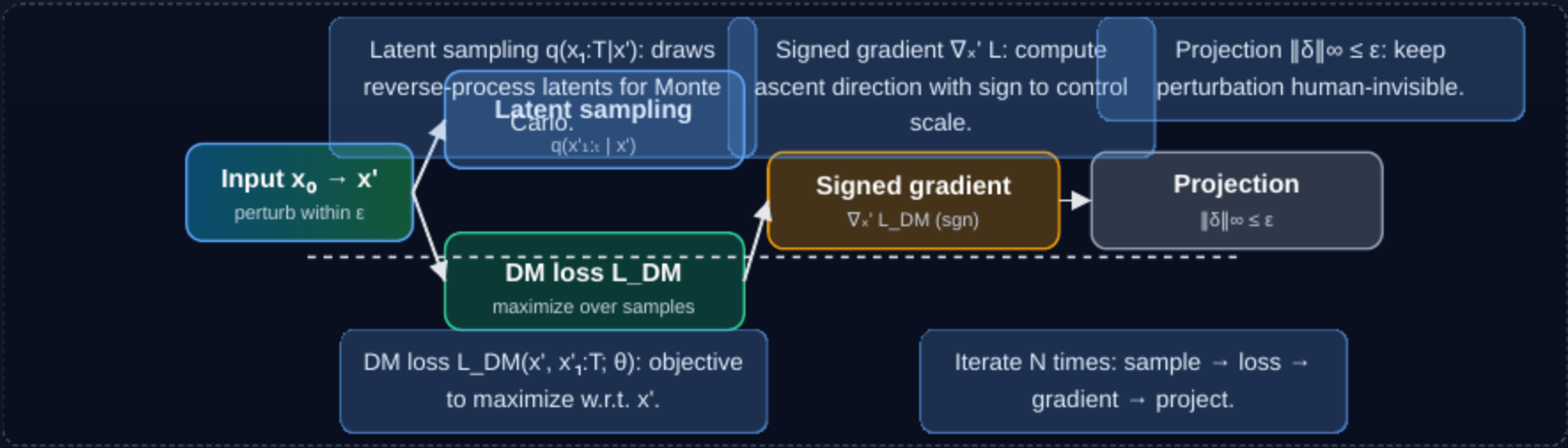
Code available

Background

- Diffusion models learn via forward noising and reverse denoising; latent diffusion performs denoising in latent space.
- Conditional generation $p_\theta(x|c)$ uses prompts, images, styles, or latent conditions.
- Adversarial examples enforce $x' = x + \delta$ under a norm bound to induce targeted behavior while keeping visuals similar.
- For DMs, the target becomes minimizing $p_\theta(x+\delta)$ to make x' out-of-distribution for the model.

Formulation: Adversarial Example for DMs

- Objective: minimize $p_{\theta}(x+\delta)$ with $\|\delta\| \leq \epsilon$ via Monte Carlo over latent trajectories $x_{1:T}$.
- They maximize the DM training loss over sampled latents to obtain δ .



Workflow: sample latents from $q(x_t; T|x')$, maximize the diffusion-model loss to get a signed gradient, project the update to an l_{∞} ball, and iterate—yielding an adversarial x' whose features are hard for the DM to extract.

AdvDM: Monte Carlo Gradient Ascent

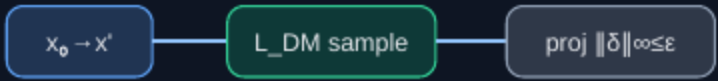
- They sample latent trajectories $x'_{1:t} \sim q(x_{1:t}|x)$ and estimate gradients of the expected DM loss.
- They perform signed gradient ascent with step size α and enforce an l_{∞} budget ϵ .

INPUTS

- Image $x_0 \rightarrow x'$ (within ϵ)
- Parameters θ , MC steps N
- Step α , budget ϵ
- (ties to Fig: x_0/x' box)
- ($q(x_t; T|x')$ node uses θ)

OUTPUT

- Adversarial image $x_{adv} = x_0 + \delta$
- (after gradient + projection nodes)



Evaluation: Protocol, Scenarios, Setup

- Protocol: two-stage inference — feature extraction $p_{\theta}(c|x)$ then generation $p_{\theta}(x|c)$; metrics: FID and Precision (Recall reported).
- Scenarios: (1) text-to-image via textual inversion (S^*), (2) style transfer via S^* with source latent, (3) image-to-image via latent z_t .
- Setup: LDM backbone; l_{∞} with 1/255 per-step, 8/255 total; 40 steps; batch 4; LSUN (cat, sheep, airplane) and WikiArt; 8× RTX A4000.

Results: Text-to-Image (Textual Inversion)

- They use 1k images/class in 5-image groups to learn S^* and generate 10k images/dataset.
- They observe large FID increases and Precision drops across LSUN classes.

LSUN-cat — FID 34.9 $\rightarrow$ 127.0

Precision 0.56 $\rightarrow$ 0.17

Generated images fail to capture target objects when trained on adversarial examples.

Most Surprising Finding

They find that tiny, human-invisible perturbations (as small as 4/255) are sufficient to prevent textual inversion from learning object/style concepts.

- Example: On LSUN-airplane, $\epsilon=4/255$ raises FID from 54.03 to 116.79 and drops Precision from 0.659 to 0.09.

Qualitative Results and Open Questions

- Style transfer: when S^* is learned from adversarial samples, generated images deviate markedly from target styles (WikiArt artists).
- Image-to-image: Stable Diffusion produces unrealistic outputs with adversarial inputs versus plausible results with clean inputs.



OPEN QUESTIONS

- How robust are AdvDM perturbations across different diffusion backbones and training variants?
- Can certified bounds be developed for minimal ϵ that guarantees failure of condition extraction?